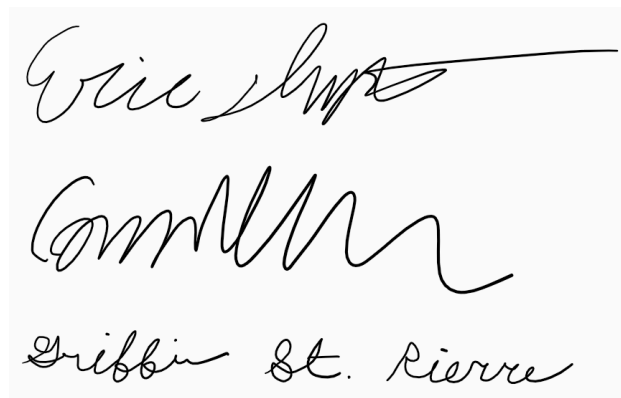


Restoring Natural Gait Cycle to Foot Drop Patients

BME 3300: Biomedical Engineering Design

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Statement of Design	CHS
Background	GMS
Objectives/Constraints Functions/User	GMS
Requirements	CHS
Preliminary Evaluations	CHS
Presentation of Final Choice	EI
Design Evaluation	EI, CHS
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Ethics	EI, CHS, GMS
Future Development	EI
Summary and Conclusion	EI
Formatting	EI

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Abstract

Drop foot is the general term for the affected step resulting from loss of function of the peroneal nerve, which leaves the patient unable to control the plantar or dorsiflexion of their foot. The cause of drop-foot is usually an underlying autoimmune, musculoskeletal, neurological, and spinal disorder (NIH, 2024). Because of the number of causes, there are many treatment options available to drop-foot patients. Surgical intervention, physical therapy, splints and braces, and powered orthotics are the primary treatments of foot drop. Some of these treatments and devices are designed for those with temporary drop-foot. Surgical interventions are primarily used for permanent drop foot cases. Through our research, we aimed to understand and develop a device that improves upon the capabilities of current market products for those with permanent drop-foot. This would entail returning the user to a more normal gait cycle that improves upon the affected gait cycle present without the device. There are two primary orthotic device categories for the treatment of drop foot. Splints and braces are the common orthotic device however they provide a limited return to normal gait cycle, their goal being to remove the tripping hazard. Powered orthotics are not a common market solution and are primarily shown in research with some success, though limited movement to the market. Our team's goal was to work on a product that more closely returns the patient to a normal gait cycle whether this be a brace or a powered orthotic. Both design types were explored with multiple active and powered devices proposed and iterated upon. Our team came to the conclusion that the active, elastic device would be the most economical and natural design. It has a low profile that can be concealed by clothing and the design added more plantar and dorsiflexion to their movement.

Executive Summary

Drop-foot has proven to be a difficult condition to effectively deal with. The leading cause of drop-foot, being nerve damage, takes a while to recover from, if recoverable. It is likely a person will develop imbalances that are difficult to correct, caused by compensatory habits forming when the damage or irritation occurs. Current affordable products have a relatively low success rate at correcting the user back to natural walking patterns due to their rigid nature. There is a need for a more active device at an effective cost primarily for people with permanent drop-foot that could benefit those with short-term drop-foot. The team brainstormed different devices that would be more active, creating various spring-based, elastic-based, and sensor driven devices. Based on the team's objectives and constraints, the elastic-based design became the top choice and pre-liminary modeling began. The elastic-based design went through 5 iterations, starting with our works-like model that was velcro and elastic bands. From there the team created multiple is-like models that utilized a 3D printed PLA brace and toe clip. Each iteration became a more discrete and sleeker model, making the brace smaller and a half-fit near the ankle rather than around the whole ankle. The elastics were replaced with bungee cords for easier demonstration purposes but a flat elastic was included in the future development model. The toe clip was made smaller in the future development design as the current one is very bulky. Ethics and regulatory issues helped guide the design and keep clear goals in mind, one being discreteness, to take the user's awareness away from the device and be hideable in societies less accepting of assistive devices. With the regulatory issues such as medicare and FDA, the current device would not pass them as more testing would be required to ensure that it meets the strict standards for assistive devices being that it's supportive to the affected area and prevents further harm from happening. Overall, the device created has potential after further testing and iterations in the hope of becoming the market standard one day.

Statement of Design Problem

The market lacks an affordable, discrete, active drop foot device that encourages recovery and allows for both plantar flexion and dorsiflexion of the foot during the gait cycle. Current devices are braces that hold up the foot to prevent tripping when walking but are insufficient for walking up stairs and slopes. To emulate a more natural gait cycle the wearer should be able to plantar flex and dorsiflex during horizontal and vertical movement. A normal gait cycle is shown in Figure 1.

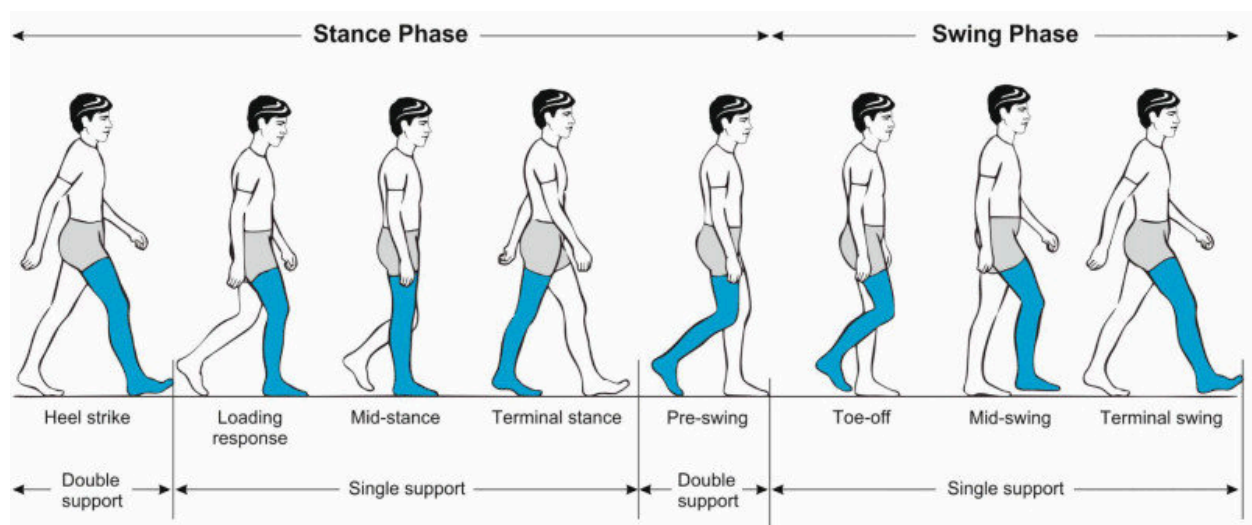


Figure 1. *Demonstration of normal gait cycle*

Background

Patients with drop-foot lack control or have little control over the movement of their ankle and as such their foot drags during the gait cycle, causing tripping in the forward swing phase without adaptation of the gait cycle. This adaptation usually includes lifting the knee and leg higher than normal, known as steppage gait, which can lead to unbalanced muscle development and potential injury. Another common adaptation is laterally swinging the foot and leg to keep the foot from striking the ground before the heel. This swing can cause hip hiking where one hip is swung higher leading to damage in the hip flexor muscle that takes on more force during walking.

The two nerves that control the lower legs, feet, and toes and provide sensation to them are the sciatic nerve and the peroneal nerve. The sciatic nerve starts between the fourth and fifth vertebrae in the spine and runs down the back of the leg. The peroneal nerve branches off the sciatic nerve near the knee and runs down the front of the leg. Any injury in the lower back, or fracture at the joints, or bones in the leg, can cause compression on the sciatic or peroneal nerve and cut off neural connections necessary to control the ankle or foot. Typically the major causes of drop-foot are any damage that occurs in the peroneal nerve or through lumbar radiculopathy which involves compression of the sciatic nerve at the fifth vertebrae. The general nerve pathing that originates from the L5 vertebrae can be seen in Figure 2.

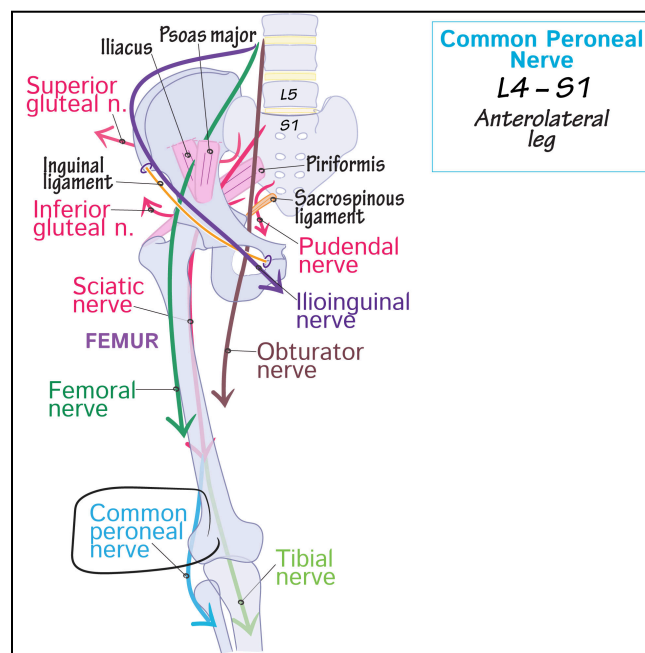


Figure 2. *Location of nerves in the lower back, hip, and leg*

Proper diagnosis of drop-foot typically involves going to a qualified medical professional for a physical exam to test the strength of the ankle in various orientations. For a more in-depth analysis, getting an electromyography(EMG) or MRI would be an appropriate course of action. An EMG involves placing specific needles around the affected area into the skin and muscles of a person. EMG will be able to detect and report the electrical activity of muscles and the change in electrical activity when a muscle is contracted.

Once a diagnosis has been made, typically treatment would be the next step. That could look like physical therapy, the use of an orthotic device, an electrical stimulation device, or surgery for more severe cases. One should proceed with less invasive recovery techniques as soon as possible to prevent the need for more invasive methods later on and prevent any irrecoverable damage from happening. Early intervention can also reduce the time needed to heal which is crucial since recovery can take a month to many months.

Physical therapy can be beneficial, by using a band to put the ankle and foot through motions like dorsiflexion, plantarflexion, and sideways movements like eversion and inversion. These movements can strengthen existing neural connections and create new neural pathways through a process called neuroplasticity, which can be identified by muscle twitching in the affected area.

For those with permanent damage to their ankle and foot muscles, these methods would be less effective. Dealing with permanent drop-foot can be difficult as long-term effects include weakness and decreased muscle mass, worsening balance, and mobility issues. This makes all the more need for a device that allows the wearer to walk as they did before developing drop-foot while promoting a more natural gait cycle.

Objectives/Constraints Functions/User

Current devices on the market work well enough for patients with drop foot, but there are some glaring flaws in their design. The biggest objective for our drop-foot device would be to have an adequate support system for the ankle and foot to correct to a more efficient and natural gait pattern. This would go hand in hand with reducing the risk of tripping and falling. To achieve these objectives, the most important constraints for our device were compared against each other.

Table 1. Pairwise Comparison Chart for the team's drop-foot device

Objective	Functionality	Cost	Comfort	Durability	Weight	Ease of Use	Total Score
Functionality	x	1	½	1	1	1	4.5
Cost	0	x	0	0	0	0	0
Comfort	½	1	x	1	1	1	4.5
Durability	0	1	0	x	½	½	2
Weight	0	1	0	½	x	½	2
Ease of Use	0	1	0	½	½	x	2

As seen in Table 1., we determined our device should prioritize functionality and comfort. The device should be comfortable to encourage use, correct a person's gait, and prevent prolonged exposure to bad walking patterns caused by drop-foot and braces. While the durability and ease of use of the device fall below functionality and comfort, they help minimize maintenance needed by professionals and maintenance done by the user. By making the device lightweight, it would reduce energy expenditure by the user and encourage use just like comfort. If these constraints are met, it would be easy to justify any additional costs, making cost the lowest priority.

With our device, it should provide sufficient support to the user's ankle. With each step, the foot should be allowed to enter dorsiflexion when in mid-air and upon touchdown, allowing the heel to hit first. Then, the foot should relax and enter plantarflexion as it pushes off and is picked up again. This core function should remain consistent when going up and down slopes and stairs. The device should work with any type of footwear and be adjustable to fit the user.

Requirements

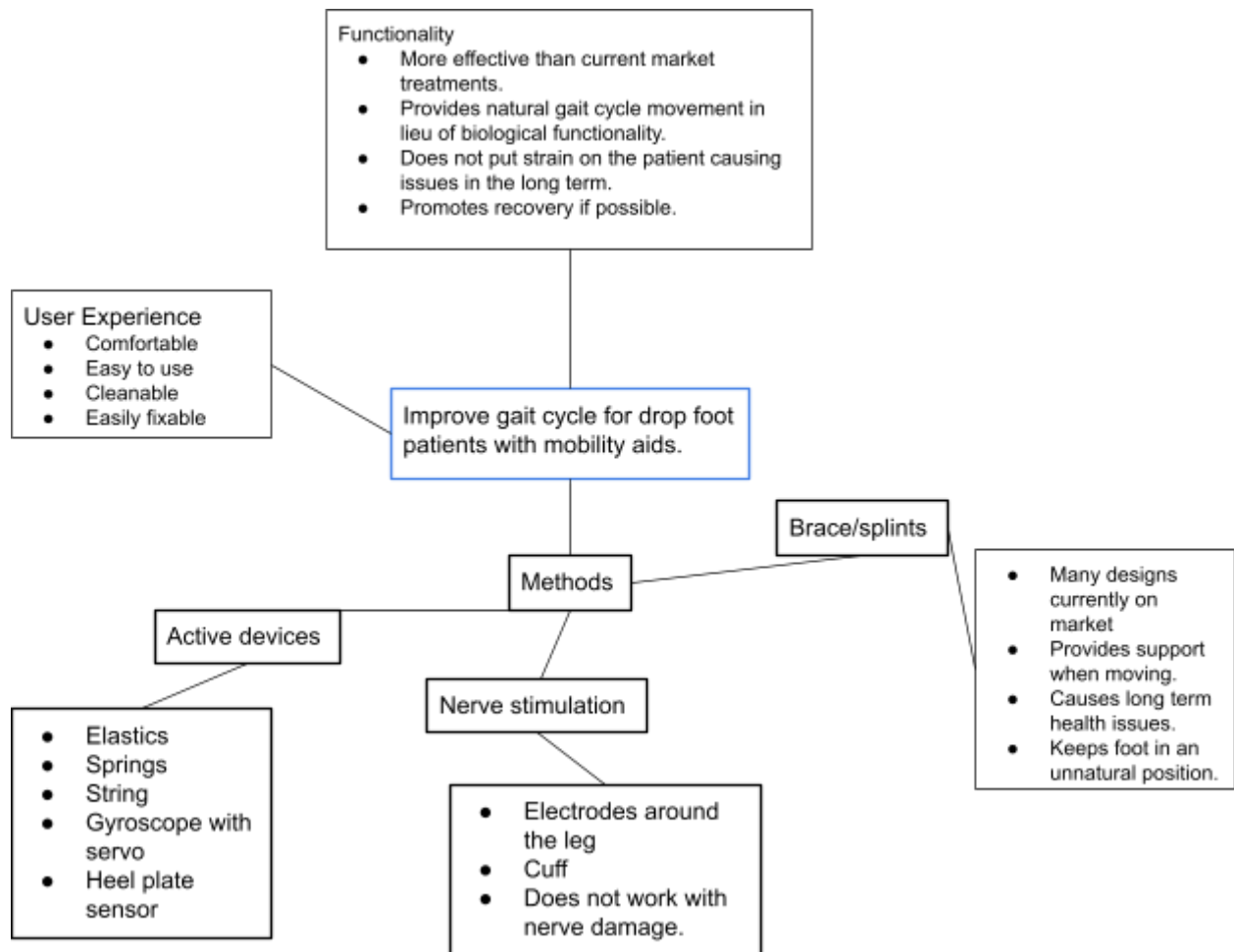


Figure 3. Concept map for the Stop Drop project

Before designing the device we had to consider what were the primary motivators for making this new design and what were some of the baseline requirements for a functional orthotic. We designed with a case of permanent drop foot in mind, for a patient with no chance of recovery through time or therapy. This allows us to design for all drop foot cases as milder cases will still have the same needs but with an eventual recovery. As shown in Figure 3 we started with a concept map to try and work out some basic ideas and some principles of different design solutions. We knew that we wanted to restore as much mobility as possible so with our research into the gait cycle we decided upon an ability to walk with the device at two miles per hour, with up to 30-50% plantarflexion and 15-30% dorsiflexion which is the norm for a working foot. This

would place the device at a higher standard than current market products as those tend to not return plantarflexion and in many cases dorsiflexion to the user. We also agreed that the device had to be similar or cheaper in price to current market alternatives or the improvements made might be inconsequential compared to a price increase. Additionally, as a wearable device this needs to be able to withstand the human body. This means a device that is antimicrobial and doesn't absorb sweat. Finally, we wanted a durable device, so that it would not break under usual pressures and be able to withstand normal environmental and weather factors. We considered these factors to be a minimum of -10 F and 120 F and when exposed to excessive sand, rain, snow, or mud the device would continue to function as intended. Whilst this is a device that will mostly be under constraint stress we still believe that the device should be able to last patients 10-15 years before they are required to purchase a new one, however with the limitations of the design time we will not be able to properly gauge its long term durability.

Alternative Solutions

A number of designs were created by members of the team to try and identify the important design decisions that would help to influence our final product. Some of these ideas shared a number of similar points.

Pneumatic Pressurized Chamber Design

The first of our proposed design was using a pneumatic tube to raise and lower the forefoot over the course of the gait cycle. A pump positioned at the sole of the foot would be inflated with some gas and connected to a tube with a lever that was attached to the toe hook. When the heel pump was compressed the gas would flow into the tube and lower the lever, thus lowering the foot. When the foot was lifted the gas would flow back into the pump and raise the toe. This would emulate the gait cycle especially during swing phase and hopefully return a more normal stance that would make ascending or descending stairs or inclines an easier task for the patient.

Corkscrew Motor with Heel Plate

This device would similarly be driven by a heel plate as with the prior device, the heel plate being released would trigger the motor to begin. It would pull the corkscrew up the shaft where it would be attached to a lever which pulls upon the cord that connects to the forefoot. The aim would be to lift the forefoot whilst the leg is in the swing phase so that it could avoid obstacles and the tension would release whilst the heel plate is compressed allowing the wearer to descend stairs and slopes.

Spring Based Tensioners

Again this device functions like the prior two with a heel plate that would activate the rest of this device. In this instance the heel plate isn't just a button but the movement of the plate causes the rest of the device to move. The heel plate is connected to a set of springs that are secured at the bottom of the shoe, so when the heel plate is compressed the springs are pushed up and the wire holding the forefoot in a lifted position are released and the forefoot is lowered. When the heel plate is released these springs contract and pull the forefoot up again so that it doesn't interfere with the swing phase.

Two Spring Design

Similar to the previously discussed design this uses the compression and release of springs to move the forefoot. The spring near to the back of the foot would control when and how the forefoot spring compressed and the forefoot spring would lift the forefoot during the swing phase. In discussing this design we made a second iteration that used another heel plate to control the movement of the back of the foot spring which would compress when the foot was lowered and expand the forefoot spring lowering the foot. When lifted the spring would expand, bringing the heel plate to its max extension and compressing the forefoot spring lifting the foot.

Heel Strike Activated Servo Actuated Foot Lifter

This idea functions similarly to the corkscrew motor but there are some primary differences. To begin there would be no 'heel plate' rather there is a sensor of some kind that identifies when heel strike occurs or identifies when the heel is in contact with the ground. When this sensor condition is met the servo would activate to unfurl some wire and let the foot go loose

when touching the ground. Plantarflexion would be allowed and when the sensor condition is not met the servo would draw back the wire and lift the forefoot.

Gyroscopic Heel Actuator

The gyroscopic heel actuator design functions quite similarly to the prior design, with a key difference. Instead of a sensor identify when heel strike occurred or when the heel was on the ground the gyroscope would identify changes in the movement of the foot and leg and the intent would be for when the gyroscope reads conditions that identify swing phase the servo actuators would pull wire of manually actuate the ankle to lift the forefoot similarly to an unaffected person. We also hoped that the gyroscope would be able to detect when the person would need to plantar flex and be able to not just lift the forefoot but also angle it downwards. This is one of the designs the team was most interested in but we had a feeling early that we would be unable to complete this design during the time constraints of the project.

Elastic Pulley

This design would be the most passive device on the list. The idea would be for a pulley/solid hardpoint that the elastic is wound around and it connects down to the foot. It would attempt to emulate normal tendons and permanently hold the foot in a raised position. This device wouldn't return as much of the normal gait cycle, but the give of the elastic would allow for the dorsiflexion needed to climb slopes or ascend stairs and with some control over the foot remaining an individual could induce plantar flexion. This device we believed to be the least clunky and we considered that it would be the least cost of all the designs.

Elastics with Moving Tensioners

This design slightly resembles the heel strike device. Some sensor would detect either when the device was in contact with the ground or when the device was lifted and in swing phase. Whichever way it went the tensioners would be on tracks and connected by a single elastic that compresses them. This would likely lift the top tensioner when the patient was in swing phase to lift the toe and lowering the top tensioner and extending the lower tensioner so that the foot was loose and in a flat position for standing. This idea received less elaboration than some of the other designs as it was a later inclusion in our design considerations.

Preliminary Evaluations

When making our final choices before moving onto the final product we began by making a chart of advantages, disadvantages, and materials for each of the list designs, alongside any extra drawings that were made for these designs. We went through as a group to evaluate which ideas had the most potential and which ideas had the most likely chance of working within the classes' timeframe.

Firstly we began to decide between some of the major ideas such as motorized or spring based or active devices. We quickly ruled out motorization as we were unsure that the device would be able to react quickly enough to the changes happening during the gait cycle. When we were evaluating whether we could make a motorized device move fast enough for our purposes we came to the conclusion that the device when moving fast enough would jerk on the patients forefoot or leg and cause discomfort or injury during use which was directly antithetical to our design constraints. We also believed that the motors, servos, and other parts would create a far heavier device than we intended and that would decrease its usefulness and make patients less likely to use it.

We pretty firmly agreed that the heel plate would be most effective for any of our activated device designs and that the primary flaw was that it would take up space beneath the foot. The other options were a sensor or a gyroscope and the sensor was believed that it would be too easily damaged by the constant contact with the ground. The gyroscope programming was beyond the scope of our knowledge so we settled on evaluating the uses of a heel plate. This idea was used to go forward with the spring based tensioners and the pneumatic pump design.

We decided upon an elastic element, secured by an ankle "shin guard" piece and attached to the forefoot by a velcro loop. This decision was made in the relative scope of the class. This met the requirement to improve upon existing market devices by allowing for dorsiflexion in all patients and allowing for plantar flexion in patients who were recovering. We also believed that the device would be more comfortable than some of the other designs as it seems lighter and less bulky. We considered under the advice of professor Page trying to fit this device inside of a shoe but as of that time the device was designed for out of shoe. Additionally falling under comfort

and ease the device seemed most likely to actually be used by patients. It was far easier to put on and off and wouldn't be a bother to someone who might hesitate to take the time to put on a bulky or complex device.

Table 2. Pugh Analysis of current drop foot devices and potential improved design ideas

Requirement	Weight	Baseline	Elastic-based	Spring-based	Gyroscope w/ servo
Functionality	5	0	1	1	1
Cost	1	-1	0	0	-1
Comfort	5	-1	1	-1	0
Durability	3	1	-1	0	0
Weight	3	-1	1	0	0
Ease of Use	3	0	1	-1	-1
Rank score		-6	13	-3	1

1 - nice to have 5 - must have *Baseline = Static drop foot brace in the \$35-\$50 range.

Presentation of Final Choice

Based upon the research and design work done by the team, we chose to pursue a semi-active, elastic-based device, branded as the Stop Drop. We chose an elastic-based design mainly due to time constraints and having a better understanding of a spring-based design over a multi-part spring or gyroscopic design. The advantages when compared to traditional drop foot devices were also apparent. By observing our Pugh Analysis chart (Figure 2), it is apparent that the elastic based device was the highest rated when compared to existing and alternative solutions. This indicates an anticipated success in implementation and overall value as a product.

Below is a technical breakdown of the device, its iterations, and future improvements and models.

The basis of our design was an active device that could lift the front of the foot to promote heel strikes during normal walking. The difference in our design from current market options is the ability to “relax” the front of foot lift to enable a more natural transfer of weight during the gait cycle. Rigid devices limit any dorsiflexion and plantar flexion, as they immobilize the foot and ankle. Similarly, semi-rigid devices commonly limit plantarflexion, with some dorsiflexion built into the design, but not achieved actively. Initial design concepts for an active drop foot device are shown below in Figures 4 and 5.

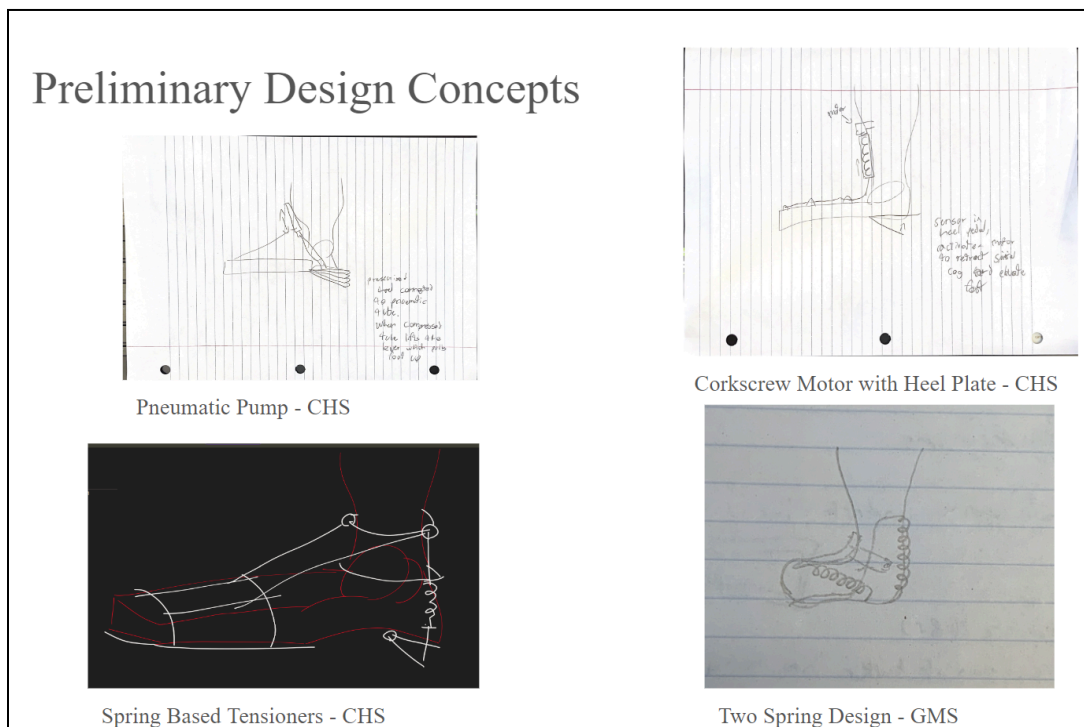


Figure 4. Initial Design Conceptual Sketches

Preliminary Designs Cont.

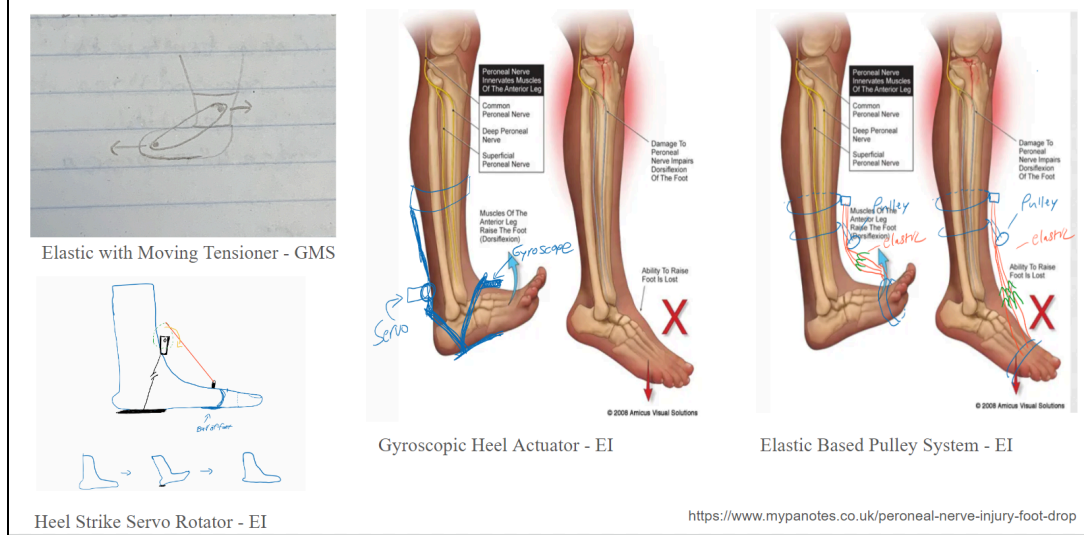


Figure 5. Initial Design Conceptual Sketches

Our design utilizes an elastic force to provide both upward support to the foot, with the allowance for downward movement without a limited range of motion. Our initial design test consisted of velcro straps and rubber elastic exercise bands. This concept was promising, as it created a dorsiflexion angle of between $\sim 15^\circ$ - 20° , which is within our design requirements. The initial concept testing can be seen in Figure 6, below.

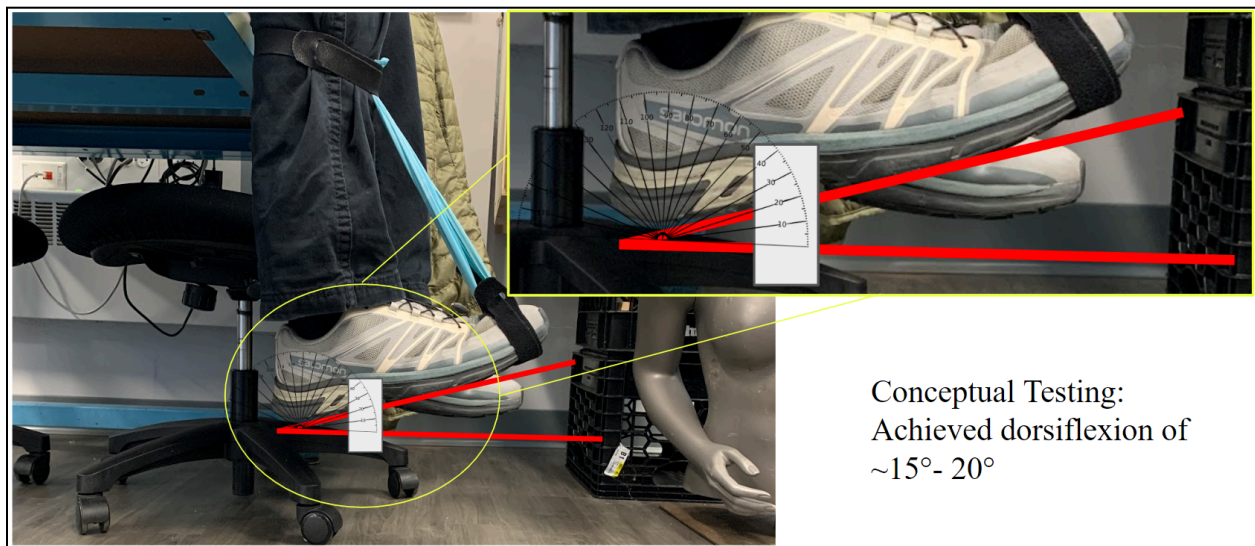


Figure 6. Initial concept testing and angle measurement.

After demonstrating the feasibility of the conceptual design, the group shifted focus to making the solution achieve all of our design goals. Since comfort was an important design consideration, making the device attach and operate discretely became a guiding direction for our future iterations

Further iteration was focused primarily on how the elastics would be attached to both the leg and the foot. Initial concepts are shown in Figure 7.

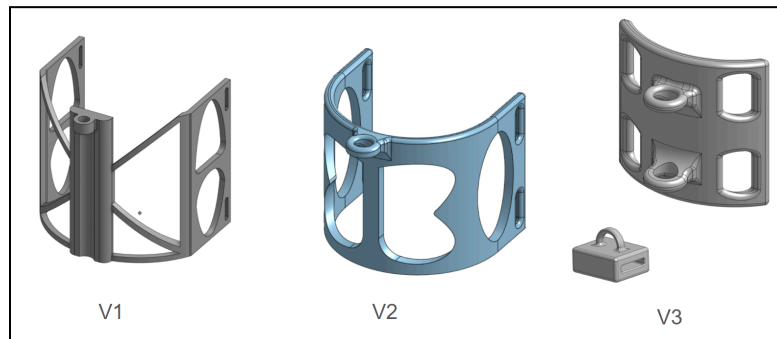


Figure 7. First three iterations of the Stop Drop shin component

Main differences between iterations V1-V2 focused on strength and elastic support to promote comfortable use. An important design choice was made from V2-V3, where the device took more of a shin guard form, rather than a “U” shape. This choice was made to accommodate a wider user population, where ankle and leg dimensions may vary. Following the V3 iteration, we printed and tested the device. It was successful in achieving the desired plantarflexion angle, while the dorsiflexion angle lacked slightly. These angles are shown in Figure 8.



Figure 8. Flexion angles of current device design

V3 provides a more comfortable, secure solution than the previous versions and allows for near universal usability, as the shin component is constructed to fit most ankle sizes and shapes. It is ~2.5" wide by ~3.5" tall with a curvature of ~2.4° to accommodate the shape of the shin. Additional antimicrobial padding sits behind the shin component to provide a barrier between the plastic (or other) material and the skin / clothes. The padding keeps the surface free from bacterial build up, with regular cleaning, and adds grip to the device for added security. Further testing and body dimension data analysis could further influence the dimensions and shape. In testing, the device worked as intended and was described by our test user as comfortable, supportive of the foot when the drop foot was simulated, and even providing some upward "spring" like force during the final phase of the gait cycle when the foot "launches" for the next step. This is encouraging feedback, as we could infer that the device would help individuals with less control of their leg / ankle.

Design Evaluation

To ensure our device will withstand normal walking conditions, we performed calculations and stress testing to determine the maximum force required to lift the foot and material limits.

Calculations for minimum force required to lift an average foot, which is approximately 2 pounds in weight and 10 inches in length, assuming a 165 pound human are shown in Figure 9.

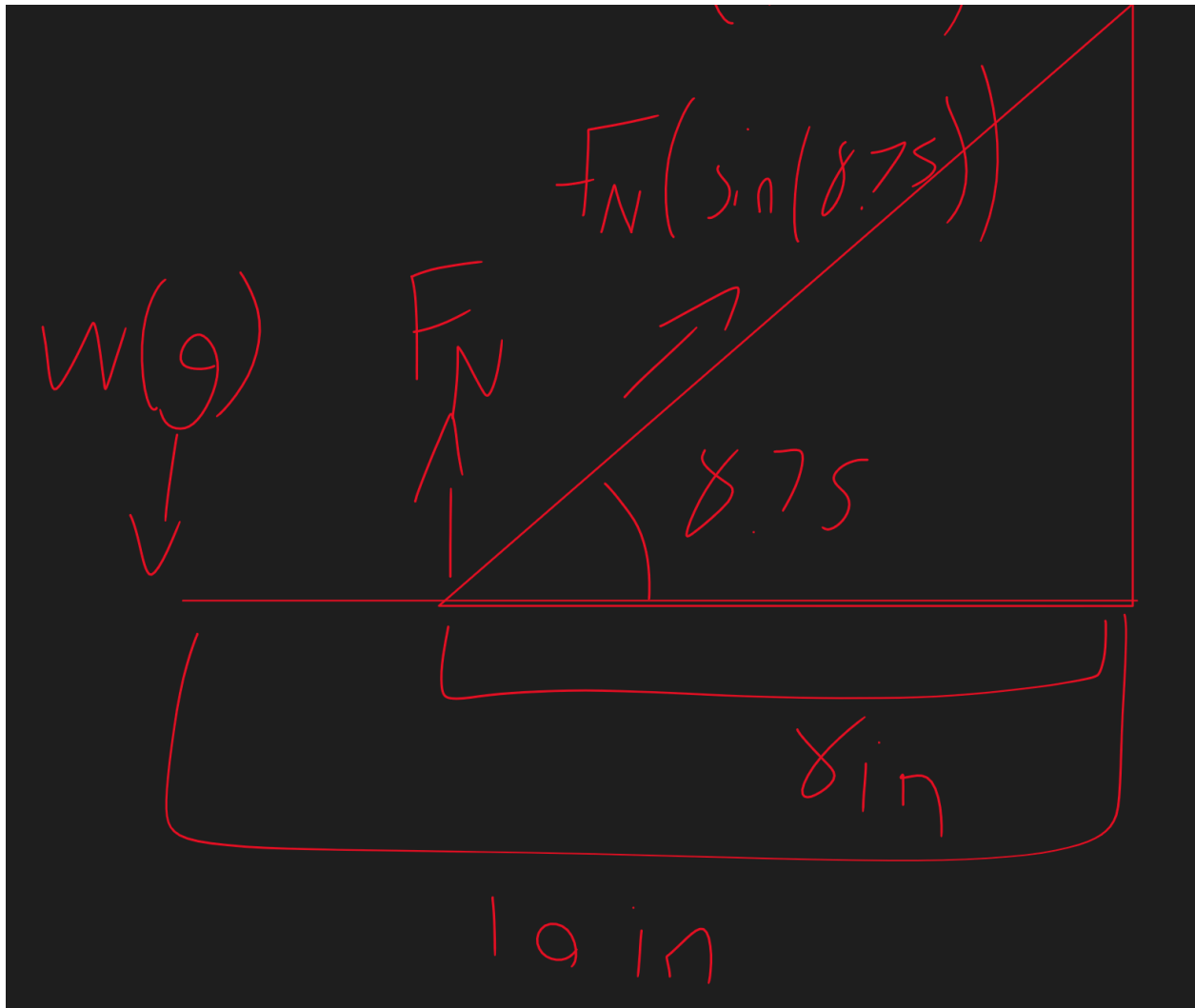


Figure 9. Calculations for minimum required force to lift the foot.

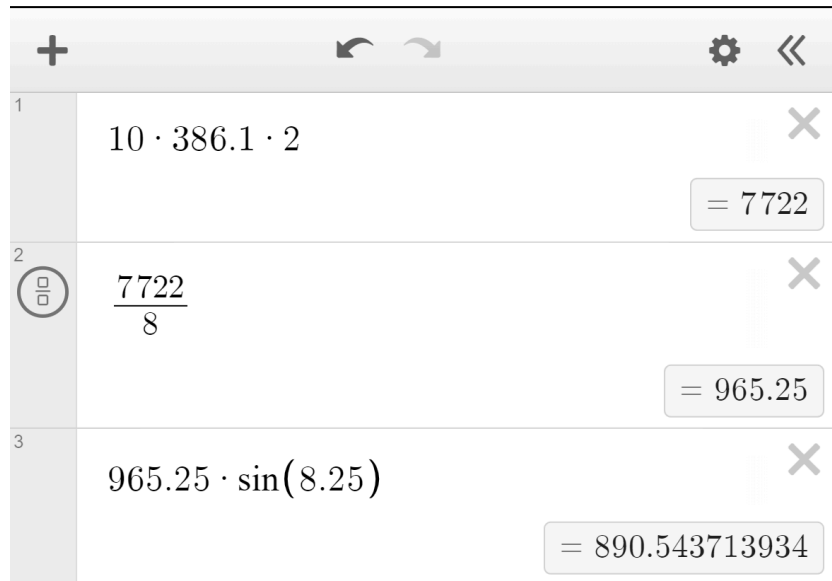


Figure 10. Calculations for minimum required force to lift the foot (in Newtons)

Our primary prototyping material, PLA, was tested using a tensile testing machine to determine the limit of its strength. A sample tensile test is shown in Figure 11 below, with the maximum pressure reaching around 650N before breaking. Across ten tensile tests, the average was 672.2 N. According to our calculations on the maximum required force to lift the foot, PLA would not be the ideal material to construct the device out of.

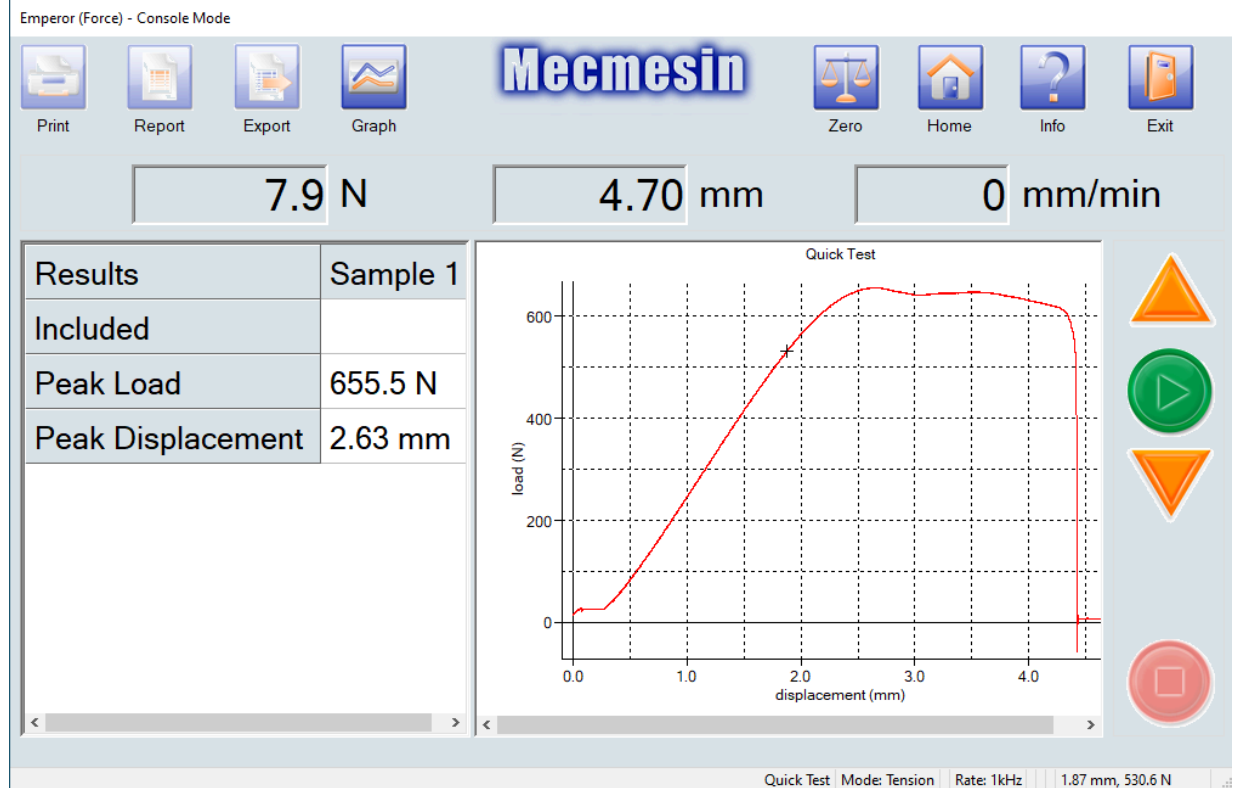


Figure 11. Tensile test for PLA plastic

Although the material withstood our prototype testing, the tensile test indicates that the device is prone to abrupt breaking, rather than withstanding the pressure or deforming slowly. Using a stronger and less brittle material would help ensure a strong, reliable device.

Regulatory Issues

Our device would most likely not be eligible for Medicare or similar coverage programs due to its non-rigid nature. To be considered eligible for coverage, the device must be sufficiently rigid as to immobilize the affected limb. According the Social Security Act §1861(s)(9) a device must be: Be rigid or semi-rigid; Support a weak or deformed body part; Restrict or eliminate motion in a diseased or injured body part; Be reasonable and necessary for the diagnosis or treatment of an illness or injury; and Improve the functioning of a malformed body part. Our device only covers a few of those standards and would fail Medicare approval. Similarly, for FDA approval, the device would need adhere to guidelines shown in [Code of Federal Regulations] [Title 21,

Volume 8] [CITE: 21CFR890.3475], which states that the device must be made “To support, to correct, or to prevent deformities or to align body structures for functional improvement”. Our device would need rigorous testing and possible redesign if we wanted to obtain FDA approval.

More on the business and manufacturing side, a standard regulation that the project should consider adhering to would be the ISO 13485 standard. This standard highlights the qualifications and tests a product must go through to be a quality medical device. While our drop-foot device is more simplistic, passing the standards set by ISO 13485 would be beneficial in covering other more specific standards. ISO 13485 could have the potential to set up our design long-term, should modifications be made to better the device. ISO 13485 is a quality management system that overlooks the creation and development of a device, how an item is marketed, treatment of the item, use of the item, and repair of an item. It enforces proper documentation of how and when something is completed during the process and who did it for proper backlogging if an issue arises. ISO 13485 would identify risks in our process of creating our drop-foot device and keep the quality and craftsmanship up to par.

Another standard regulation that would be important for the team to implement would be ISO 14001. ISO 14001 is largely based on the processes for a product and its impact on the environment. By complying with this standard, the impact on the environment from a product would fall between negligible to non-existent. A company that takes its impact on the environment seriously and actively tries to reduce it shows good faith and promotes public faith in the company. If consumers dislike a company that can have a negative impact with them either not using the product, or even protesting against that company, which will in the long run, run a company out of business.

Financials

This device was designed using standard, off the shelf parts apart from the custom 3D printed components. Of the components used in our testing phase, most were not deemed medical grade, nor do they necessarily need to be, as the device is worn outside of the body and any interface directly with the body is mitigated with antimicrobial padding material. This helps lower the overall cost and hassle when procuring the materials. Figure 9 displays a rough breakdown of the cost to produce a single Stop Drop device using the currently sourced materials.

Part, Raw Materials, Components	Vendor	Unit Cost	Quantity	Total Cost
Elastic / Bungee	Amazon	\$3.00	1	\$3.00
Velcro Strapping	Amazon	\$1.00	3	\$3.00
Anti Microbial Padding	Amazon	\$13.00	1	\$13.00
PLA Plastic	Matterhackers	\$20	0.05	\$1
			Total	\$20.00

Figure 12. BOM for the Stop Drop device materials

Ethics

There are likely fewer ethical considerations that need to be made with an assistive device like ours. With our device, we would need to anticipate cultural aspects of the device like some places not believing in assistive devices and acceptance of them. Effective marketing to spin them as not assistive devices would be very helpful to gain acceptance to communities more hesitant or against the use of our design. Making the device more subtle by allowing the wearer to conceal it as much as possible would promote the buying and use of our product.

Another consideration is our device should not have any risk associated with it such as causing pain to the user or affecting their gait cycle more. To prevent this from happening, extensive testing of our design through stress tests, fatigue tests, and speed tests should be done to make sure it is durable and effective. These tests would allow us to adjust our designs to be more durable, while ensuring proper function.

Future Development

The main focus for future development of the Stop Drop is design towards a lower profile device. A low profile device would be more attractive for users, as the impact of their personal image or self esteem would be reduced, allowing the user to carry out daily functions without judgment. Below, future plans for modification are shown in Figures 13, 14, and 15. The primary change from V3 to the future device is the move from bungee cord to a flat elastic band. The foot attachment point will also be moved into the shoe with a lower profile clip.

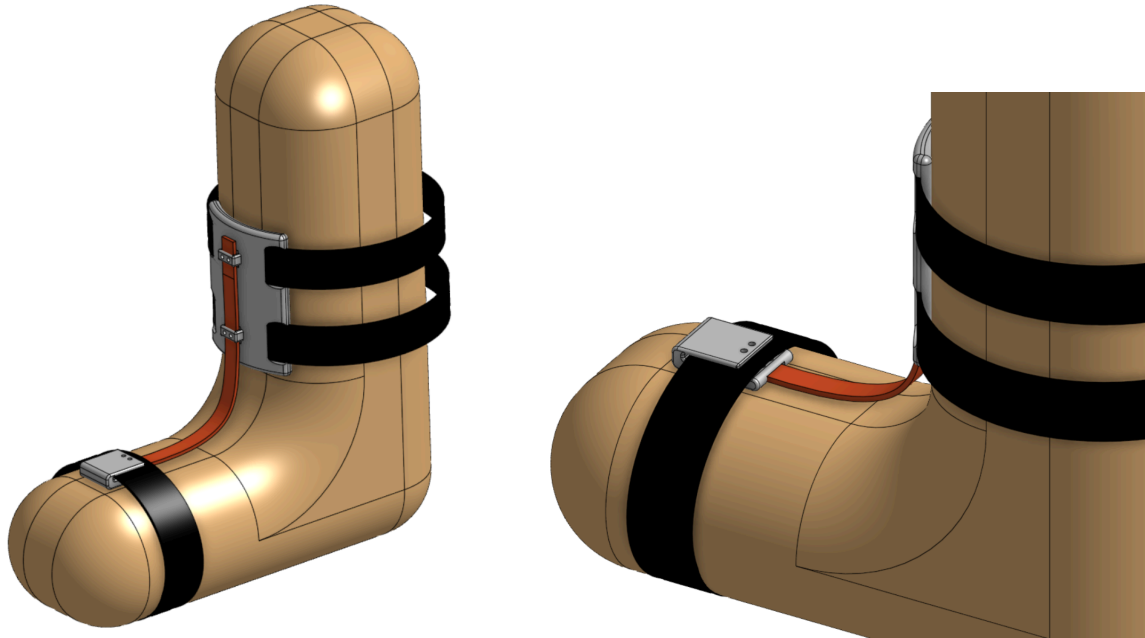


Figure 13. CAD model of the future iteration of the Stop Drop.

This model represents the next iteration of our current design. A major design change is the foot attachment point moving inside the shoe. This allows for a more discrete wearing experience while maintaining functionality. An advantage of this design compared to current devices is the lack of shoe modification. Technical drawings of the V4 design shown below in Figures 14 and 15.

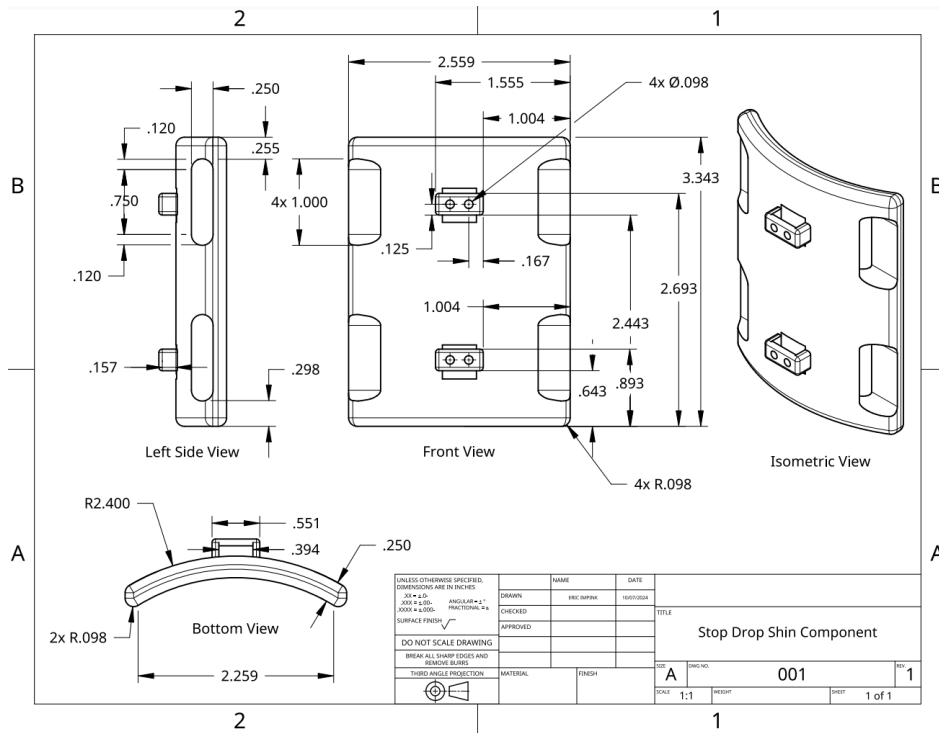


Figure 14. Technical drawing of the Stop Drop shin component

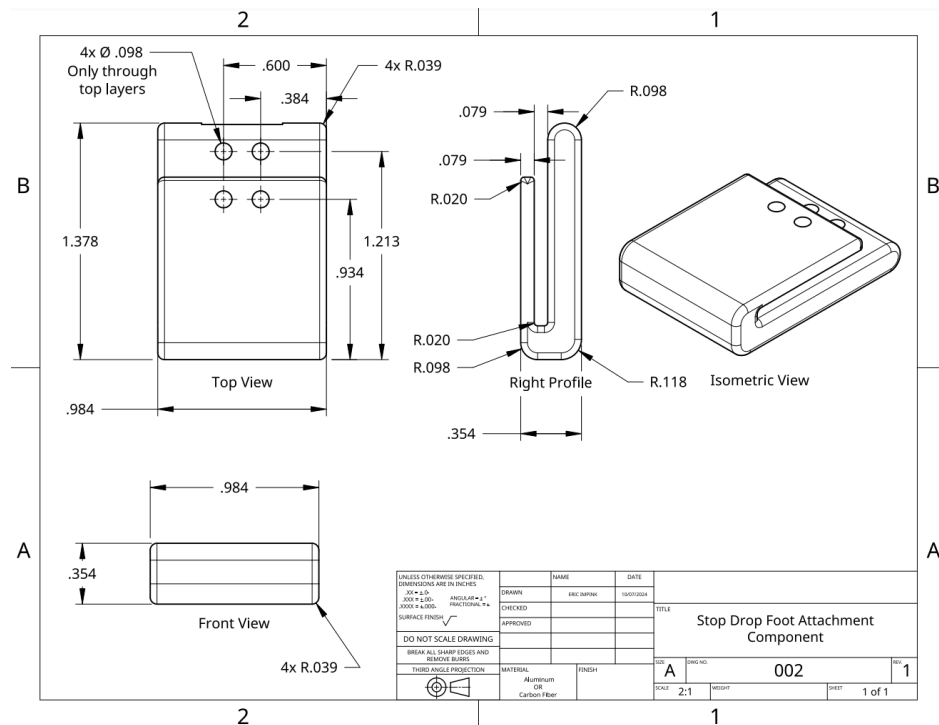


Figure 15. Technical drawing of the Stop Drop foot/sock component

Device function will need to be re-evaluated with the future modifications. We have not tested with a 1cm wide elastic band, nor have we tested attachment inside of the shoe. The shin component remains mostly unchanged, with the attachment points changing from circular holes to rectangular attachment slots.

Summary and Conclusion

By designing with affordability in mind, the device is constructed using a majority of off the shelf parts that are globally available. This allows for implementation in nearly all settings where commercial stores / materials are present. It also allows for easy implementation to places of need, as the overall cost of the device is low, creating an opportunity for a widespread impact on low-income communities as well as third world country settings.

In terms of the effectiveness, our testing proved that our device was successful in improving the gait cycle, when compared to the simulated drop foot gait cycle. The elastic design improves over current market products of a similar design by increasing the range of plantar and dorsiflexion whilst still maintaining support during the swing phase. Additionally, the device will not require bulky, exoskeleton-like support from outside the shoe and will remain nearly invisible when worn under pant cuffs.

The Stop Drop is a promising evolution in drop foot device technology. While the concept is not necessarily novel, it is innovative, approaching the problem of drop foot from another effective goal. Our device provides function while remaining relatively low-cost and low-profile. If the opportunity for further development and funding was available, pursuing an actuated device could prove to be an interesting avenue of discovery. The main advantages of our device is its affordability, effectiveness, and comfort.

References

- Błażkiewicz, C, Wiszomirska, I, Kaczmarczyk, K, Wit, A. (2016). *Mechanisms of compensation in the gait of patients with drop foot*, Clinical Biomechanics, Volume 42, 2017, Pages 14-19, ISSN 0268-0033, <https://doi.org/10.1016/j.clinbiomech.2016.12.014>.
- Nordqvist, C. (2019). Foot Drop. In StatPearls. StatPearls Publishing. Retrieved from <https://www.ncbi.nlm.nih.gov/books/NBK554393/>
- Alam, M., Choudhury, I. A., & Bin Mamat, A. (2014). Mechanism and design analysis of articulated ankle foot orthoses for drop-foot. *TheScientificWorldJournal*, 2014, 867869 .<https://doi.org/10.1155/2014/867869>
- Cleveland Clinic. “Foot Drop: What It Is, Causes, Symptoms & Treatment.” Cleveland Clinic, <https://my.clevelandclinic.org/health/symptoms/17814-foot-drop>.
- Kafri, Michal, et al. “Falls in Stroke Patients: Risk Factors and Risk Management.” International Journal of Stroke, vol. 10, no. 1, 2015, pp. 92-98. PubMed Central, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4994968/>.
- Hickey, Robert, et al. “Durability of Deep Transcranial Magnetic Stimulation for Veterans with Treatment-Resistant Depression with Comorbid Suicide Risk and PTSD Symptoms.” Psychiatry Research, vol. 332, 2024, <https://www.sciencedirect.com/science/article/pii/S0268003316302340>.

Appendices

Appendix A: Questions for O&P Clinics

1. How familiar are you with drop foot?
2. How does drop foot affect the lives of your patients?
3. What issues do you believe exist with existing devices?
4. What issues do your patients express with existing devices?
5. What feature(if any) do you believe would improve patient recovery or quality of life?
6. Do your patients express a preference to a device that could be worn within a shoe or without?

Appendix B: Performance Specification

1. Performance
 - Sustain a travel speed of 2 mph
 - Dorsiflex between 15-30° and plantarflex between 20-50°
 - Provide just above the minimal amount of force to lift the front of the foot
2. Comfort and Usability
 - Comfortable to encouraging use
 - Light as possible (2 lbs) to minimize energy expenditure
 - Easy to put on and take off
 - Wearable for long periods (8-10 hours)
 - Non-irritable
3. Noise level
 - Operate quietly (30-40 dB) to minimize disturbance
 - Not exceed 60 dB to disturb walking conversation
4. Durability and maintenance
 - Weather-resistant
 - Manufacturable and fit for mass production
 - Require little maintenance

- Maintain a natural gait cycle
- Cleanable
- Adjustable

Appendix C: User Guide

1. Align the provided velcro loop with the ball of your foot, which is usually the widest part of the foot at the front. Secure the velcro loop around your shoe and attach the foot clip to the top part of the velcro loop, on the TOP of the shoe.
2. Put on the shin component on the leg and align it so the bottom of the shin component rests just above the heel part of the shoe. Secure the shin component with the provided velcro straps.
3. Attach the elastic from the shin component to the toe component using the provided elastic and attachment points. The device should pull the foot to a near 90° angle.
4. Verify all velcro straps are secure.
5. The device is ready for use!

For cleaning:

Remove the device from the body and clean it with a chemically light disinfectant spray. Allow the device components to air dry. For additional cleaning, remove the shin pad and clean it with dish soap. Allow for complete air drying before reinstalling.